

Memo 03: MP1 Mass-Conservation Check and Updated Hypotheses

FVCOM-MP Delaware Test Cases

April 29, 2026

Purpose

This memo records the latest 10-day comparison of cases **a**, **b**, and **c**, and updates the diagnostic hypotheses after the recent improvement to the split-field `conc/cnew` update before processor exchange. The main target is the null-floc comparison: `conc` in case **a** should match `conc_d` in case **b** when the kinetic interaction terms and aggregation fractions are zero.

Current Evidence

All three extracted MAT files now cover the same model-day window, FVCOM day 58485.0–58494.957031, or comparison day 0.00–9.96. The updated Python comparison uses this shared time origin.

Case	Records	Final water (kg)	Final bed (kg)	Final total (kg)	Retained vs. a (%)
a	240	1001.25	0.00	1001.25	100.0
b	240	371.85	0.00	371.85	37.1
c	240	504.26	24.29	528.55	52.8

The split-field exchange/recombination change recovered substantial mass in case **b** relative to the earlier comparison, but it did not close the budget. Case **b** still ends 629.40 kg below case **a**; case **c** ends 472.70 kg below case **a**. In case **b**, all retained mass is in `mp1_dis`; `mp1_agg` remains zero in the saved output, which is the expected null-kinetics behavior.

Cap and Split Checks

The hard cap is no longer a leading explanation for this 10-day test. The largest extracted concentrations are far below the code cap of 100 kg m^{-3} :

$$\max C_a = 4.43 \times 10^{-4}, \quad \max C_b = 6.98 \times 10^{-3}, \quad \max C_c = 1.45 \times 10^{-2} \text{ kg m}^{-3}.$$

The cap-threshold counters are zero for all split variables in cases **b** and **c**. The cap code remains present in `mod_plast.F:1242--1243`, but the present output indicates it is not active.

The saved split fields are internally consistent:

$$\text{mp1_agg} + \text{mp1_dis} \approx \text{mp1}.$$

The maximum total split-minus-combined mismatch is zero for case **b** and $1.83 \times 10^{-7} \text{ kg}$ for case **c**. Therefore the saved-output recombination is not the present sink.

Coding Symmetry Check

The static setup is mostly symmetric. In `init_plast.F:200--208`, `INI_CONC_AG_RATE=0` initializes `conc_a = 0`, `conc_d = conc`, and matching `cnew_a/cnew_d`. The generic plastic input files also match except for `PLAST_FLOC`: source aggregation, initial aggregation, mass aggregation, and OBC aggregation are all zero.

The main remaining code-path differences are:

1. **Runtime namelist asymmetry.** Case `a` uses `SEDIMENT_MODEL=F`, while case `b` uses `SEDIMENT_MODEL=T`. This means the two cases are not only `PLAST_FLOC=F` versus `PLAST_FLOC=T`; they also enter different compiled/runtime sediment branches.
2. **Transport is one combined tracer versus two split tracers.** Case `a` calls `adv_scal` once on `conc` (`mod_plast.F:1261--1267`). Case `b` calls `adv_scal` first on `conc_a` and then on `conc_d` (`mod_plast.F:1166--1183`). With `SOURCE_AG_RATE=0`, the `conc_d` source should equal the combined source, but the split path still performs an extra zero-tracer pass for `conc_a`.
3. **Vertical diffusion is also split.** The `floc` path calls `vdif_scal` on `cnew_a` and `cnew_d` (`mod_plast.F:1202--1204`); the `no-floc` path calls it once on `cnew` (`mod_plast.F:1276--1278`). In exact arithmetic this should be equivalent when `cnew_a=0`, but it is still a checkpoint for roundoff, wet/dry, and halo-state differences.
4. **Open-boundary and point-source boundary conditions are split.** The `floc` path applies OBC and point-source boundary calls separately to `conc_a/conc_d` (`mod_plast.F:1543--1554` and `mod_plast.F:1600--1605`); the `no-floc` path applies them once to `conc` (`mod_plast.F:1563--1568` and `mod_plast.F:1612--1614`). Because `OBC_AG_RATE=0` and `SOURCE_AG_RATE=0`, `conc_d` should receive the full boundary/source value, but this remains a leading place to compare post-operator mass.
5. **The first MPI exchange still exchanges combined cnew in the split branch.** Before OBC handling, the `floc` branch exchanges `cnew_a`, `cnew_d`, and `cnew` (`mod_plast.F:1513--1516`). At that point the meaningful updated state is split; combined `cnew` is not rebuilt until `mod_plast.F:1715--1716`. This may be harmless if no code consumes combined `cnew` in between, but it is not strictly symmetric and should be cleaned or checkpointed.
6. **Final exchange/recombination was improved but should remain an invariant check.** The split path now clips `cnew_a/cnew_d`, exchanges those same arrays, then rebuilds `conc = cnew = conc_a + conc_d` (`mod_plast.F:1657--1716`). The mass recovery in case `b` supports this as a real previous asymmetry.
7. **Bed/surface update is algebraically different but inactive for case b.** The `floc` branch splits erosion and surface input by `ratio_bottom/ratio_surface` and applies a 10x growth limiter (`mod_plast.F:2383--2440`); the `no-floc` branch adds directly to `conc` (`mod_plast.F:2451--2461`). However, the extracted case-a/b bed mass and sampled dep/ero integrals are zero, so this is not the leading explanation for the current a/b gap.

Updated Hypotheses

1. **Leading hypothesis: transport/OBC/source operator asymmetry.** Since the cap is inactive and bed exchange is zero for case **b**, the remaining mass deficit most likely appears between `adv_scal`, `vdif_scal`, `bcond_scal_OBC`, `bcond_scal_PTsource`, and the final non-negative clamp/exchange. A checkpoint budget after each of these operators should show the first divergence.
2. **Configuration asymmetry must be isolated.** A true symmetry test should compare no-floc and null-floc runs with the same `SEDIMENT_MODEL` setting. The cleanest next pair is an `a_sedT` case using `PLAST_FLOC=F` and `SEDIMENT_MODEL=T` versus the current null-floc case **b**.
3. **Hard cap is demoted for this run.** The cap remains a real non-conservative code site, but current maxima and threshold counts show it is not active in the present 10-day outputs.
4. **Nonnegative clipping remains possible.** The current extraction tracks cap counts and negative saved mass, but it does not measure mass removed internally by `max(...,0)` before output. This is still a plausible silent sink if the split transport path produces transient negative undershoots that are clipped before the saved time slice.
5. **Bed-exchange and erosion/surface limiters are secondary for the case-a/b mismatch.** They remain important for case **c** and full floc interaction, but the present null-floc case **b** has zero final bed mass and zero sampled dep/ero exchange.

Focused Next Test

1. Run a matched `SEDIMENT_MODEL=T`, `PLAST_FLOC=F` baseline. If this aligns with case **b**, the current difference is mostly a sediment-runtime comparison issue. If it aligns with case **a**, the split path remains the cause.
2. Add a short internal checkpoint diagnostic for one small run. Record domain-integrated mass for `conc`, `conc_d`, and `conc_a + conc_d` after: start, deposition, erosion/surface update, `adv_scal`, `vdif_scal`, OBC, point-source BC, nonnegative clamp, MPI exchange, and recombination.
3. For the same diagnostic, explicitly accumulate mass removed by nonnegative clipping and the hard cap. The cap should remain zero in this test; any nonzero clipping loss would be immediately informative.
4. If serial and parallel runs differ, focus on the two exchange locations: `mod_plast.F:1513--1516` before OBC and `mod_plast.F:1677--1716` at the final split recombination.